

NorthStar

In-House Goal Setting & Tracking Portal

AtomQuest Hackathon 1.0 Submission

Live Demo	onenorthstar.vercel.app
Frontend Repository	github.com/iamnitishsah/NorthStar-Frontend
Backend Repository	github.com/iamnitishsah/NorthStar-Backend

Developed By

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Login Credentials

Credentials	Employee	Manager	Admin
ID	EMP0001	MGR1101	ADMIN
Password	pass123	pass123	pass123

Submitted for AtomQuest Hackathon 1.0

1 Project Overview

NorthStar is a structured web-based Goal Setting & Tracking Portal designed to replace fragmented spreadsheet-driven workflows with a centralized digital system.

The platform manages the complete lifecycle of employee goals:

- Goal drafting and submission
- Manager approval workflow
- Goal locking and admin unlock flow
- Quarterly achievement tracking
- Shared departmental KPIs
- Audit logging and governance
- Analytics and reporting

The system provides dedicated experiences for:

- Employee
- Manager (L1)
- Admin / HR

2 Technology Stack

Layer	Technology
Frontend	React, TypeScript, Tailwind CSS
Backend	FastAPI, Python, Pydantic
Database	MongoDB
Authentication	JWT Authentication
Background Tasks	Celery + Redis
Charts	Recharts
Deployment	Vercel + Render + MongoDB Atlas

3 System Architecture

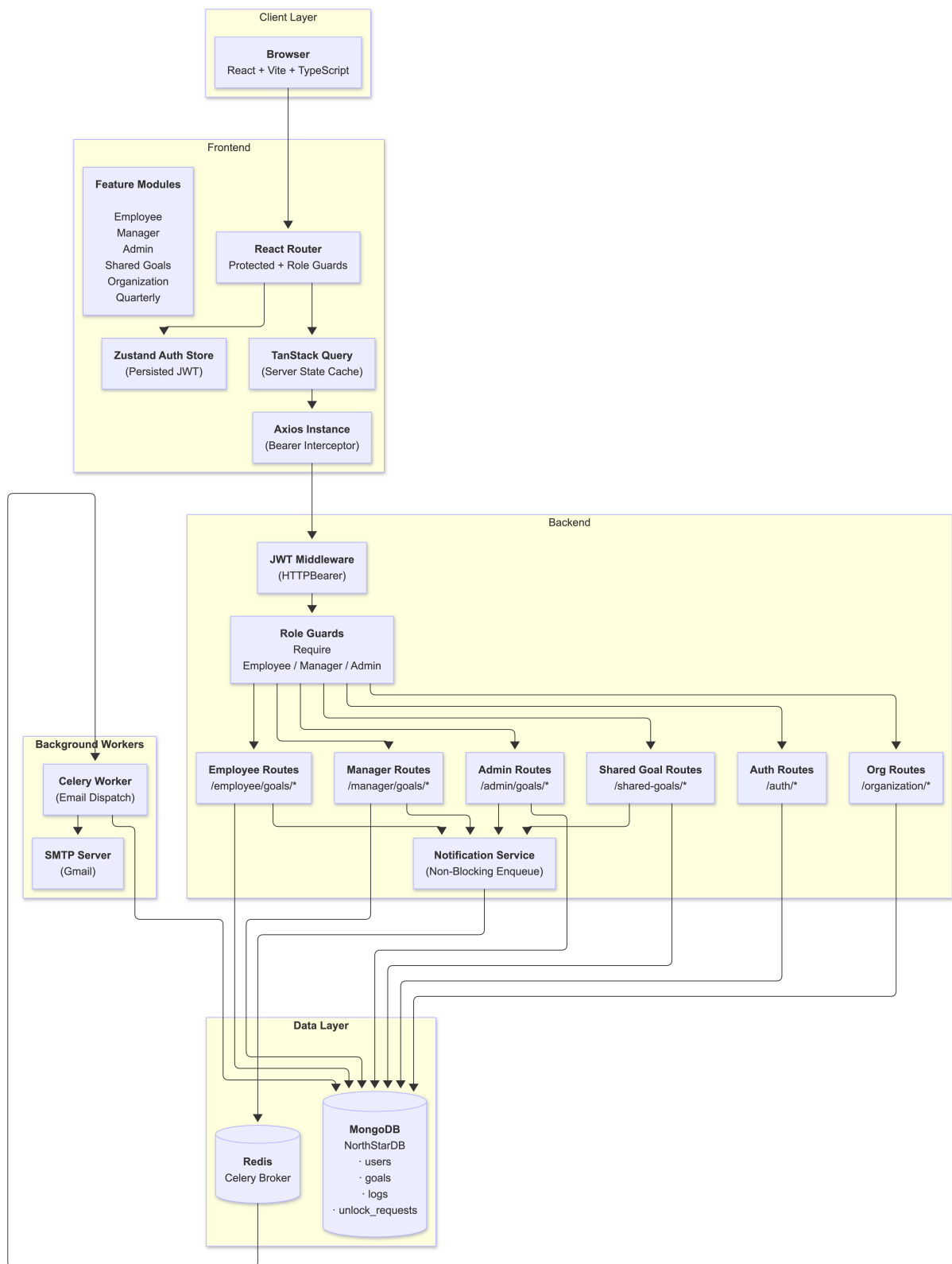


Figure 1: NorthStar System Architecture

4 Core Workflow

4.1 Goal Creation & Approval

- Employees create goals using:
 - Thrust Area
 - Goal Title
 - Description
 - UoM Type
 - Target Value
 - Weightage
- Employees submit goals for manager review
- Managers can:
 - Approve goals
 - Return goals for correction
 - Edit target and weightage inline
- Approved goals move to locked state
- Locked goals can only be modified through Admin Unlock flow

4.2 Quarterly Achievement Tracking

- Employees submit quarterly achievement updates
- Managers review planned vs actual progress
- Managers add structured check-in comments
- Progress percentage is computed automatically

4.3 Shared Goals

- Managers or Admins can assign shared KPIs to multiple employees
- Shared goals support:
 - Editable weightage
 - Read-only targets and title
 - Centralized achievement synchronization

5 User Roles

Role	Capabilities
Employee	Create goals, edit drafts, submit goals, quarterly check-ins, track progress
Manager	Review submissions, approve or return goals, add check-in comments
Admin / HR	Unlock goals, manage reports, audit logs, analytics, shared goals

6 Goal Lifecycle

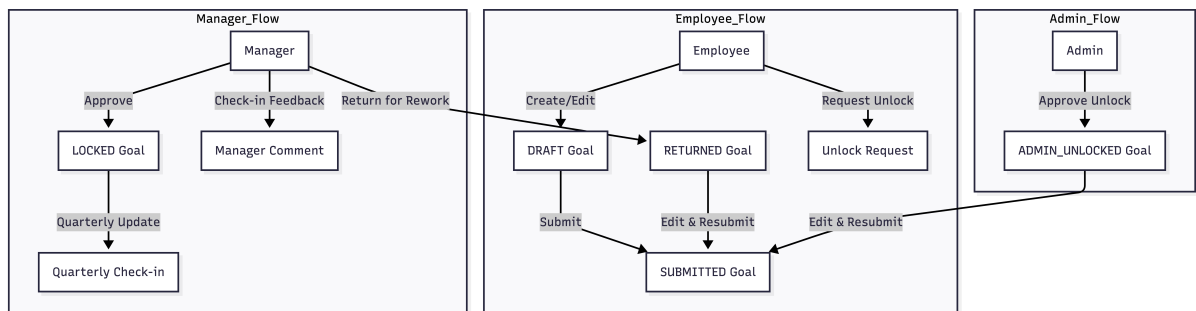


Figure 2: Goal Lifecycle Workflow

State	Description
DRAFT	Goal created by employee. Editable and deletable.
SUBMITTED	Submitted for manager review. Waiting for approval or return.
RETURNED	Returned by manager for correction and resubmission.
LOCKED	Approved by manager. Goal becomes immutable for employees.
ADMIN_UNLOCKED	Unlocked by Admin after approval. Employee can edit and resubmit again.

Unlock Flow

- Employees can request unlock for locked goals
- Admin reviews unlock requests
- Approved unlock requests move goals to **ADMIN_UNLOCKED** state
- Employees can modify and resubmit unlocked goals
- All unlock actions are recorded in audit logs

7 Validation Rules

- Total goal weightage must equal **100%**
- Minimum weightage per goal: **10%**
- Maximum goals per employee: **8**
- Only approved (locked) goals allow quarterly check-ins
- Shared goal targets remain immutable for recipients

8 Progress Score Computation

UoM Type	Logic	Formula
Min (Numeric / %)	Higher is better	Achievement $\div$ Target
Max (Numeric / %)	Lower is better	Target $\div$ Achievement
Timeline	Deadline based	Completion Date vs Deadline
Zero Based	Zero means success	If 0 $\rightarrow$ 100%, else 0%

9 Key Features

- JWT-based authentication and role-based access control
- Goal approval workflow with inline manager editing
- Quarterly check-in and progress tracking
- Shared departmental KPIs with sync support
- Audit logging for governance and traceability
- Analytics dashboards and reporting modules
- Background email notifications using Celery + Redis
- Responsive UI with role-specific dashboards

10 Deployment

Component	Hosting
Frontend	Vercel
Backend API	Render
Database	MongoDB Atlas
Task Queue / Cache	Render Redis